algorithms lend themselves to analysis using recurrences, so do parallel algorithms in the fork-join model.

- The fork-join programming model is faithful to how multicore programming has been evolving in practice. A growing number of multicore environments support one variant or another of fork-join parallel programming, including Cilk [290, 291, 383, 396], Habanero-Java [466], the Java Fork-Join Framework [279], OpenMP [81], Task Parallel Library [289], Threading Building Blocks [376], and X10 [82].

Section 26.1 introduces parallel pseudocode, shows how the execution of a task-parallel computation can be modeled as a directed acyclic graph, and presents the metrics of work, span, and parallelism, which you can use to analyze parallel algorithms. Section 26.2 investigates how to multiply matrices in parallel, and Section 26.3 tackles the tougher problem of designing an efficient parallel merge sort.

26.1 The basics of fork-join parallelism

Our exploration of parallel programming begins with the problem of computing Fibonacci numbers recursively in parallel. We'll look at a straightforward serial Fibonacci calculation, which, although inefficient, serves as a good illustration of how to express parallelism in pseudocode.

Recall that the Fibonacci numbers are defined by equation (3.31) on page 69:

$$F_i = \begin{cases} 0 & \text{if } i = 0\,, \\ 1 & \text{if } i = 1\,, \\ F_{i-1} + F_{i-2} & \text{if } i \geq 2\,. \end{cases}$$

To calculate the nth Fibonacci number recursively, you could use the ordinary serial algorithm in the procedure FIB on the facing page. You would not really want to compute large Fibonacci numbers this way,

because this computation does needless repeated work, but parallelizing it can be instructive.

```
FIB (n)
1 if n ≤ 1
2     return n
3 else x = FIB (n − 1)
4     y = FIB (n − 2)
5     return x + y
```

To analyze this algorithm, let $T(n)$ denote the running time of FIB (n). Since FIB (n) contains two recursive calls plus a constant amount of extra work, we obtain the recurrence

$$T(n) = T(n-1) + T(n-2) + \Theta(1).$$

This recurrence has solution $T(n) = \Theta(F_n)$, which we can establish by using the substitution method (see Section 4.3). To show that $T(n) = O(F_n)$, we'll adopt the inductive hypothesis that $T(n) \le aF_n - b$, where $a > 1$ and $b > 0$ are constants. Substituting, we obtain

$$\begin{aligned} T(n) &\le (aF_{n-1} - b) + (aF_{n-2} - b) + \Theta(1) \\ &= a(F_{n-1} + F_{n-2}) - 2b + \Theta(1) \\ &\le aF_n - b, \end{aligned}$$

if we choose b large enough to dominate the upper-bound constant in the $\Theta(1)$ term. We can then choose a large enough to upper-bound the $\Theta(1)$ base case for small n. To show that $T(n) = \Omega(F_n)$, we use the inductive hypothesis $T(n) \ge aF_n - b$. Substituting and following reasoning similar to the asymptotic upper-bound argument, we establish this hypothesis by choosing b smaller than the lower-bound constant in the $\Theta(1)$ term and a small enough to lower-bound the $\Theta(1)$ base case for small n. Theorem 3.1 on page 56 then establishes that T

$(n) = \Theta(F_n)$, as desired. Since $F_n = \Theta(\phi^n)$, where $\phi = (1 + \sqrt{5})/2$ is the golden ratio, by equation (3.34) on page 69, it follows that

$$T(n) = \Theta(\phi^n) . \tag{26.1}$$

Thus this procedure is a particularly slow way to compute Fibonacci numbers, since it runs in exponential time. (See Problem 31-3 on page 954 for faster ways.)

Let's see why the algorithm is inefficient. Figure 26.1 shows the tree of recursive procedure instances created when computing F_6 with the FIB procedure. The call to FIB(6) recursively calls FIB(5) and then FIB(4). But, the call to FIB(5) also results in a call to FIB(4). Both instances of FIB(4) return the same result ($F_4 = 3$). Since the FIB procedure does not memoize (recall the definition of "memoize" from page 368), the second call to FIB(4) replicates the work that the first call performs, which is wasteful.

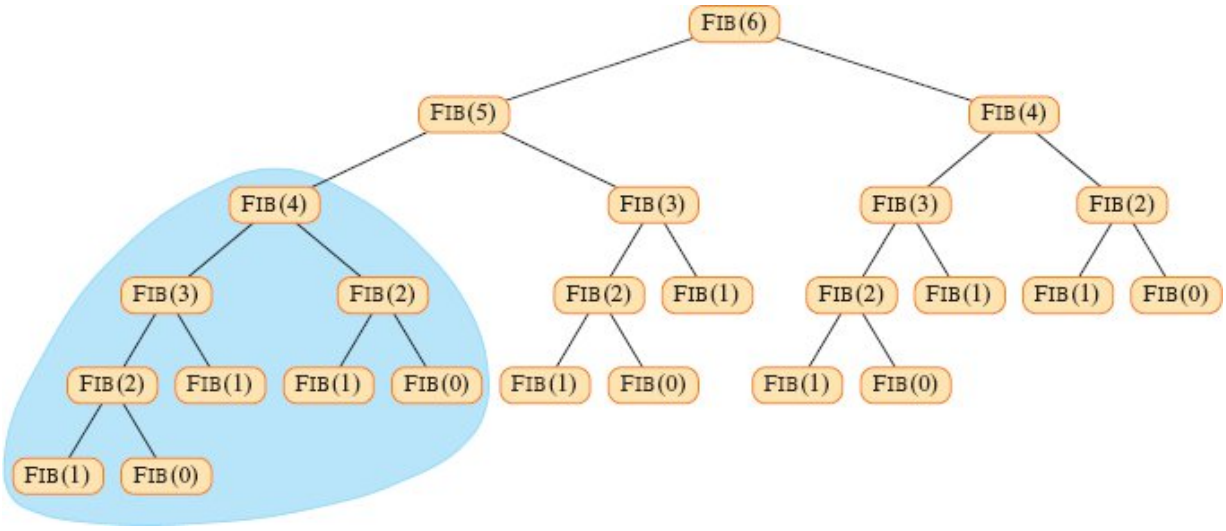


Figure 26.1 The invocation tree for FIB(6). Each node in the tree represents a procedure instance whose children are the procedure instances it calls during its execution. Since each instance of FIB with the same argument does the same work to produce the same result, the inefficiency of this algorithm for computing the Fibonacci numbers can be seen by the vast number of repeated calls to compute the same thing. The portion of the tree shaded blue appears in task-parallel form in Figure 26.2.

Although the FIB procedure is a poor way to compute Fibonacci numbers, it can help us warm up to parallelism concepts. Perhaps the

most basic concept is to understand is that if two parallel tasks operate on entirely different data, then—absent other interference—they each produce the same outcomes when executed at the same time as when they run serially one after the other. Within FIB (n), for example, the two recursive calls in line 3 to FIB ($n - 1$) and in line 4 to FIB ($n - 2$) can safely execute in parallel because the computation performed by one in no way affects the other.

Parallel keywords

The P-FIB procedure on the next page computes Fibonacci numbers, but using the ***parallel keywords*** **spawn** and **sync** to indicate parallelism in the pseudocode.

If the keywords **spawn** and **sync** are deleted from P-FIB, the resulting pseudocode text is identical to FIB (other than renaming the procedure in the header and in the two recursive calls). We define the ***serial projection***[1] of a parallel algorithm to be the serial algorithm that results from ignoring the parallel directives, which in this case can be done by omitting the keywords **spawn** and **sync**. For **parallel for** loops, which we'll see later on, we omit the keyword **parallel**. Indeed, our parallel pseudocode possesses the elegant property that its serial projection is always ordinary serial pseudocode to solve the same problem.

```
P-FIB (n)
1 if n ≤ 1
2     return n
3 else x = spawn P-FIB (n − 1)  // don't wait for subroutine to return
4     y = P-FIB (n − 2)         // in parallel with spawned subroutine
5     sync                      // wait for spawned subroutine to finish
6     return x + y
```

Semantics of parallel keywords

Spawning occurs when the keyword **spawn** precedes a procedure call, as in line 3 of P-FIB. The semantics of a spawn differs from an ordinary procedure call in that the procedure instance that executes the spawn—the ***parent***—may continue to execute in parallel with the spawned subroutine—its ***child***—instead of waiting for the child to finish, as would happen in a serial execution. In this case, while the spawned child is computing P-FIB ($n - 1$), the parent may go on to compute P-FIB ($n-2$) in line 4 in parallel with the spawned child. Since the P-FIB procedure is recursive, these two subroutine calls themselves create nested parallelism, as do their children, thereby creating a potentially vast tree of subcomputations, all executing in parallel.

The keyword **spawn** does not say, however, that a procedure *must* execute in parallel with its spawned children, only that it *may*. The parallel keywords express the ***logical parallelism*** of the computation, indicating which parts of the computation may proceed in parallel. At runtime, it is up to a ***scheduler*** to determine which subcomputations actually run in parallel by assigning them to available processors as the computation unfolds. We'll discuss the theory behind task-parallel schedulers shortly (on page 759).

A procedure cannot safely use the values returned by its spawned children until after it executes a **sync** statement, as in line 5. The keyword **sync** indicates that the procedure must wait as necessary for all its spawned children to finish before proceeding to the statement after the **sync**—the "join" of a fork-join parallel computation. The P-FIB procedure requires a **sync** before the **return** statement in line 6 to avoid the anomaly that would occur if x and y were summed before P-FIB ($n - 1$) had finished and its return value had been assigned to x. In addition to explicit join synchronization provided by the **sync** statement, it is convenient to assume that every procedure executes a **sync** implicitly before it returns, thus ensuring that all children finish before their parent finishes.

A graph model for parallel execution

It helps to view the execution of a parallel computation—the dynamic stream of runtime instructions executed by processors under the

direction of a parallel program—as a directed acyclic graph $G = (V, E)$, called a ***(parallel) trace***.[2] Conceptually, the vertices in V are executed instructions, and the edges in E represent dependencies between instructions, where $(u, v) \in E$ means that the parallel program required instruction u to execute before instruction v.

It's sometimes inconvenient, especially if we want to focus on the parallel structure of a computation, for a vertex of a trace to represent only one executed instruction. Consequently, if a chain of instructions contains no parallel or procedural control (no **spawn**, **sync**, procedure call, or **return**—via either an explicit **return** statement or the return that happens implicitly upon reaching the end of a procedure), we group the entire chain into a single ***strand***. As an example, Figure 26.2 shows the trace that results from computing P-FIB(4) in the portion of Figure 26.1 shaded blue. Strands do not include instructions that involve parallel or procedural control. These control dependencies must be represented as edges in the trace.

When a parent procedure calls a child, the trace contains an edge (u, v) from the strand u in the parent that executes the call to the first strand v of the spawned child, as illustrated in Figure 26.2 by the edge from the orange strand in P-FIB(4) to the blue strand in P-FIB(2). When the last strand v' in the child returns, the trace contains an edge (v', u') to the strand u', where u' is the successor strand of u in the parent, as with the edge from the white strand in P-FIB(2) to the white strand in P-FIB(4).

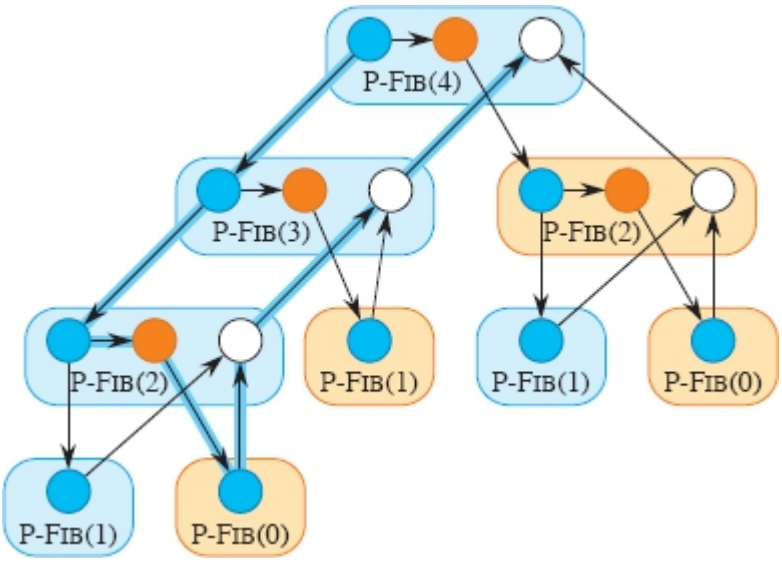


Figure 26.2 The trace of P-FIB(4) corresponding to the shaded portion of Figure 26.1. Each circle represents one strand, with blue circles representing any instructions executed in the part of the procedure (instance) up to the spawn of P-FIB ($n - 1$) in line 3; orange circles representing the instructions executed in the part of the procedure that calls P-FIB ($n - 2$) in line 4 up to the **sync** in line 5, where it suspends until the spawn of P-FIB ($n - 1$) returns; and white circles representing the instructions executed in the part of the procedure after the **sync**, where it sums x and y, up to the point where it returns the result. Strands belonging to the same procedure are grouped into a rounded rectangle, blue for spawned procedures and tan for called procedures. Assuming that each strand takes unit time, the work is 17 time units, since there are 17 strands, and the span is 8 time units, since the critical path—shown with blue edges—contains 8 strands.

When the parent spawns a child, however, the trace is a little different. The edge (u, v) goes from parent to child as with a call, such as the edge from the blue strand in P-FIB(4) to the blue strand in P-FIB(3), but the trace contains another edge (u, u') as well, indicating that u's successor strand u' can continue to execute while v is executing. The edge from the blue strand in P-FIB(4) to the orange strand in P-FIB(4) illustrates one such edge. As with a call, there is an edge from the last strand v' in the child, but with a spawn, it no longer goes to u's successor. Instead, the edge is (v', x), where x is the strand immediately following the **sync** in the parent that ensures that the child has finished, as with the edge from the white strand in P-FIB(3) to the white strand in P-FIB(4).

You can figure out what parallel control created a particular trace. If a strand has two successors, one of them must have been spawned, and if a strand has multiple predecessors, the predecessors joined because of a **sync** statement. Thus, in the general case, the set V forms the set of strands, and the set E of directed edges represents dependencies between strands induced by parallel and procedural control. If G contains a directed path from strand u to strand v, we say that the two strands are ***(logically) in series***. If there is no path in G either from u to v or from v to u, the strands are ***(logically) in parallel***.

A fork-join parallel trace can be pictured as a dag of strands embedded in an ***invocation tree*** of procedure instances. For example, Figure 26.1 shows the invocation tree for FIB(6), which also serves as the invocation tree for P-FIB(6), the edges between procedure instances now representing either calls or spawns. Figure 26.2 zooms in on the subtree that is shaded blue, showing the strands that constitute each procedure instance in P-FIB(4). All directed edges connecting strands run either within a procedure or along undirected edges of the invocation tree in Figure 26.1. (More general task-parallel traces that are not fork-join traces may contain some directed edges that do not run along the undirected tree edges.)

Our analyses generally assume that parallel algorithms execute on an ***ideal parallel computer***, which consists of a set of processors and a ***sequentially consistent*** shared memory. To understand sequential consistency, you first need to know that memory is accessed by ***load instructions***, which copy data from a location in the memory to a register within a processor, and by ***store instructions***, which copy data from a processor register to a location in the memory. A single line of pseudocode can entail several such instructions. For example, the line $x = y + z$ could result in load instructions to fetch each of y and z from memory into a processor, an instruction to add them together inside the processor, and a store instruction to place the result x back into memory. In a parallel computer, several processors might need to load or store at the same time. Sequential consistency means that even if multiple processors attempt to access the memory simultaneously, the shared memory behaves as if exactly one instruction from one of the

processors is executed at a time, even though the actual transfer of data may happen at the same time. It is as if the instructions were executed one at a time sequentially according to some global linear order among all the processors that preserves the individual orders in which each processor executes its own instructions.

For task-parallel computations, which are scheduled onto processors automatically by a runtime system, the sequentially consistent shared memory behaves as if a parallel computation's executed instructions were executed one by one in the order of a topological sort (see Section 20.4) of its trace. That is, you can reason about the execution by imagining that the individual instructions (not generally the strands, which may aggregate many instructions) are interleaved in some linear order that preserves the partial order of the trace. Depending on scheduling, the linear order could vary from one run of the program to the next, but the behavior of any execution is always as if the instructions executed serially in a linear order consistent with the dependencies within the trace.

In addition to making assumptions about semantics, the ideal parallel-computer model makes some performance assumptions. Specifically, it assumes that each processor in the machine has equal computing power, and it ignores the cost of scheduling. Although this last assumption may sound optimistic, it turns out that for algorithms with sufficient "parallelism" (a term we'll define precisely a little later), the overhead of scheduling is generally minimal in practice.

Performance measures

We can gauge the theoretical efficiency of a task-parallel algorithm using ***work/span analysis***, which is based on two metrics: "work" and "span." The ***work*** of a task-parallel computation is the total time to execute the entire computation on one processor. In other words, the work is the sum of the times taken by each of the strands. If each strand takes unit time, the work is just the number of vertices in the trace. The ***span*** is the fastest possible time to execute the computation on an unlimited number of processors, which corresponds to the sum of the times taken by the strands along a longest path in the trace, where

"longest" means that each strand is weighted by its execution time. Such a longest path is called the ***critical path*** of the trace, and thus the span is the weight of the longest (weighted) path in the trace. (Section 22.2, pages 617–619 shows how to find a critical path in a dag $G = (V, E)$ in $\Theta(V + E)$ time.) For a trace in which each strand takes unit time, the span equals the number of strands on the critical path. For example, the trace of Figure 26.2 has 17 vertices in all and 8 vertices on its critical path, so that if each strand takes unit time, its work is 17 time units and its span is 8 time units.

The actual running time of a task-parallel computation depends not only on its work and its span, but also on how many processors are available and how the scheduler allocates strands to processors. To denote the running time of a task-parallel computation on P processors, we subscript by P. For example, we might denote the running time of an algorithm on P processors by T_P. The work is the running time on a single processor, or T_1. The span is the running time if we could run each strand on its own processor—in other words, if we had an unlimited number of processors—and so we denote the span by T_∞.

The work and span provide lower bounds on the running time T_P of a task-parallel computation on P processors:

- In one step, an ideal parallel computer with P processors can do at most P units of work, and thus in T_P time, it can perform at most $P\,T_P$ work. Since the total work to do is T_1, we have $P\,T_P \geq T_1$. Dividing by P yields the ***work law***:

$$T_P \geq T_1/P \ . \tag{26.2}$$

- A P-processor ideal parallel computer cannot run any faster than a machine with an unlimited number of processors. Looked at another way, a machine with an unlimited number of processors can emulate a P-processor machine by using just P of its processors. Thus, the ***span law*** follows:

$$T_P \geq T_\infty \ . \tag{26.3}$$

We define the ***speedup*** of a computation on P processors by the ratio T_1/T_P, which says how many times faster the computation runs on P processors than on one processor. By the work law, we have $T_P \geq T_1/P$, which implies that $T_1/T_P \leq P$. Thus, the speedup on a P-processor ideal parallel computer can be at most P. When the speedup is linear in the number of processors, that is, when $T_1/T_P = \Theta(P)$, the computation exhibits ***linear speedup***. ***Perfect linear speedup*** occurs when $T_1/T_P = P$.

The ratio T_1/T_∞ of the work to the span gives the ***parallelism*** of the parallel computation. We can view the parallelism from three perspectives. As a ratio, the parallelism denotes the average amount of work that can be performed in parallel for each step along the critical path. As an upper bound, the parallelism gives the maximum possible speedup that can be achieved on any number of processors. Perhaps most important, the parallelism provides a limit on the possibility of attaining perfect linear speedup. Specifically, once the number of processors exceeds the parallelism, the computation cannot possibly achieve perfect linear speedup. To see this last point, suppose that $P > T_1/T_\infty$, in which case the span law implies that the speedup satisfies $T_1/T_P \leq T_1/T_\infty < P$. Moreover, if the number P of processors in the ideal parallel computer greatly exceeds the parallelism—that is, if $P \gg T_1/T_\infty$—then $T_1/T_P \ll P$, so that the speedup is much less than the number of processors. In other words, if the number of processors exceeds the parallelism, adding even more processors makes the speedup less perfect.

As an example, consider the computation P-FIB(4) in Figure 26.2, and assume that each strand takes unit time. Since the work is $T_1 = 17$ and the span is $T_\infty = 8$, the parallelism is $T_1/T_\infty = 17/8 = 2.125$. Consequently, achieving much more than double the performance is impossible, no matter how many processors execute the computation. For larger input sizes, however, we'll see that P-FIB (n) exhibits substantial parallelism.

We define the ***(parallel) slackness*** of a task-parallel computation executed on an ideal parallel computer with P processors to be the ratio

$(T_1/T_\infty)/P = T_1/(P\,T_\infty)$, which is the factor by which the parallelism of the computation exceeds the number of processors in the machine. Restating the bounds on speedup, if the slackness is less than 1, perfect linear speedup is impossible, because $T_1/(P\,T_\infty) < 1$ and the span law imply that $T_1/T_P \geq T_1/T_\infty > P$. Indeed, as the slackness decreases from 1 and approaches 0, the speedup of the computation diverges further and further from perfect linear speedup. If the slackness is less than 1, additional parallelism in an algorithm can have a great impact on its execution efficiency. If the slackness is greater than 1, however, the work per processor is the limiting constraint. We'll see that as the slackness increases from 1, a good scheduler can achieve closer and closer to perfect linear speedup. But once the slackness is much greater than 1, the advantage of additional parallelism shows diminishing returns.

Scheduling

Good performance depends on more than just minimizing the work and span. The strands must also be scheduled efficiently onto the processors of the parallel machine. Our fork-join parallel-programming model provides no way for a programmer to specify which strands to execute on which processors. Instead, we rely on the runtime system's scheduler to map the dynamically unfolding computation to individual processors. In practice, the scheduler maps the strands to static threads, and the operating system schedules the threads on the processors themselves. But this extra level of indirection is unnecessary for our understanding of scheduling. We can just imagine that the scheduler maps strands to processors directly.

A task-parallel scheduler must schedule the computation without knowing in advance when procedures will be spawned or when they will finish—that is, it must operate ***online***. Moreover, a good scheduler operates in a distributed fashion, where the threads implementing the scheduler cooperate to load-balance the computation. Provably good online, distributed schedulers exist, but analyzing them is complicated. Instead, to keep our analysis simple, we'll consider an online ***centralized***

scheduler that knows the global state of the computation at any moment.

In particular, we'll analyze ***greedy schedulers***, which assign as many strands to processors as possible in each time step, never leaving a processor idle if there is work that can be done. We'll classify each step of a greedy scheduler as follows:

- ***Complete step***: At least P strands are ***ready*** to execute, meaning that all strands on which they depend have finished execution. A greedy scheduler assigns any P of the ready strands to the processors, completely utilizing all the processor resources.
- ***Incomplete step***: Fewer than P strands are ready to execute. A greedy scheduler assigns each ready strand to its own processor, leaving some processors idle for the step, but executing all the ready strands.

The work law tells us that the fastest running time T_P that we can hope for on P processors must be at least T_1/P. The span law tells us that the fastest possible running time must be at least T_∞. The following theorem shows that greedy scheduling is provably good in that it achieves the sum of these two lower bounds as an upper bound.

Theorem 26.1
On an ideal parallel computer with P processors, a greedy scheduler executes a task-parallel computation with work T_1 and span T_∞ in time

$$T_P \leq T_1/P + T_\infty \,. \tag{26.4}$$

Proof Without loss of generality, assume that each strand takes unit time. (If necessary, replace each longer strand by a chain of unit-time strands.) We'll consider complete and incomplete steps separately.

In each complete step, the P processors together perform a total of P work. Thus, if the number of complete steps is k, the total work executing all the complete steps is kP. Since the greedy scheduler doesn't execute any strand more than once and only T_1 work needs to

be performed, it follows that $kP \leq T_1$, from which we can conclude that the number k of complete steps is at most T_1/P.

Now, let's consider an incomplete step. Let G be the trace for the entire computation, let G' be the subtrace of G that has yet to be executed at the start of the incomplete step, and let G'' be the subtrace remaining to be executed after the incomplete step. Consider the set R of strands that are ready at the beginning of the incomplete step, where $|R| < P$. By definition, if a strand is ready, all its predecessors in trace G have executed. Thus the predecessors of strands in R do not belong to G'. A longest path in G' must necessarily start at a strand in R, since every other strand in G' has a predecessor and thus could not start a longest path. Because the greedy scheduler executes all ready strands during the incomplete step, the strands of G'' are exactly those in G' minus the strands in R. Consequently, the length of a longest path in G'' must be 1 less than the length of a longest path in G'. In other words, every incomplete step decreases the span of the trace remaining to be executed by 1. Hence, the number of incomplete steps can be at most T_∞.

Since each step is either complete or incomplete, the theorem follows.

■

The following corollary shows that a greedy scheduler always performs well.

Corollary 26.2
The running time T_P of any task-parallel computation scheduled by a greedy scheduler on a P-processor ideal parallel computer is within a factor of 2 of optimal.

Proof Let T^*_P be the running time produced by an optimal scheduler on a machine with P processors, and let T_1 and T_∞ be the work and span of the computation, respectively. Since the work and span laws—inequalities (26.2) and (26.3)—give $T^*_P \geq \max\{T_1/P, T_\infty\}$, Theorem 26.1 implies that

$$
\begin{aligned}
T_P &\le T_1/P + T_\infty \\
&\le 2 \cdot \max\{T_1/P, T_\infty\} \\
&\le 2T_P^* .
\end{aligned}
$$

■

The next corollary shows that, in fact, a greedy scheduler achieves near-perfect linear speedup on any task-parallel computation as the slackness grows.

Corollary 26.3

Let T_P be the running time of a task-parallel computation produced by a greedy scheduler on an ideal parallel computer with P processors, and let T_1 and T_∞ be the work and span of the computation, respectively. Then, if $P \ll T_1/T_\infty$, or equivalently, the parallel slackness is much greater than 1, we have $T_P \approx T_1/P$, a speedup of approximately P.

Proof If we suppose that $P \ll T_1/T_\infty$, then it follows that $T_\infty \ll T_1/P$, and hence Theorem 26.1 gives $T_P \le T_1/P + T_\infty \approx T_1/P$. Since the work law (26.2) dictates that $T_P \ge T_1/P$, we conclude that $T_P \approx T_1/P$, which is a speedup of $T_1/T_P \approx P$.

■

The $\ll$ symbol denotes "much less," but how much is "much less"? As a rule of thumb, a slackness of at least 10—that is, 10 times more parallelism than processors—generally suffices to achieve good speedup. Then, the span term in the greedy bound, inequality (26.4), is less than 10% of the work-per-processor term, which is good enough for most engineering situations. For example, if a computation runs on only 10 or 100 processors, it doesn't make sense to value parallelism of, say 1,000,000, over parallelism of 10,000, even with the factor of 100 difference. As Problem 26-2 shows, sometimes reducing extreme parallelism yields algorithms that are better with respect to other concerns and which still scale up well on reasonable numbers of processors.

Analyzing parallel algorithms

We now have all the tools we need to analyze parallel algorithms using work/span analysis, allowing us to bound an algorithm's running time on any number of processors. Analyzing the work is relatively straightforward, since it amounts to nothing more than analyzing the running time of an ordinary serial algorithm, namely, the serial projection of the parallel algorithm. You should already be familiar with analyzing work, since that is what most of this textbook is about! Analyzing the span is the new thing that parallelism engenders, but it's generally no harder once you get the hang of it. Let's investigate the basic ideas using the P-FIB program.

Analyzing the work $T_1(n)$ of P-FIB (n) poses no hurdles, because we've already done it. The serial projection of P-FIB is effectively the original FIB procedure, and hence, we have $T_1(n) = T(n) = \Theta(\phi^n)$ from equation (26.1).

Figure 26.3 illustrates how to analyze the span. If two traces are joined in series, their spans add to form the span of their composition, whereas if they are joined in parallel, the span of their composition is the maximum of the spans of the two traces. As it turns out, the trace of any fork-join parallel computation can be built up from single strands by series-parallel composition.

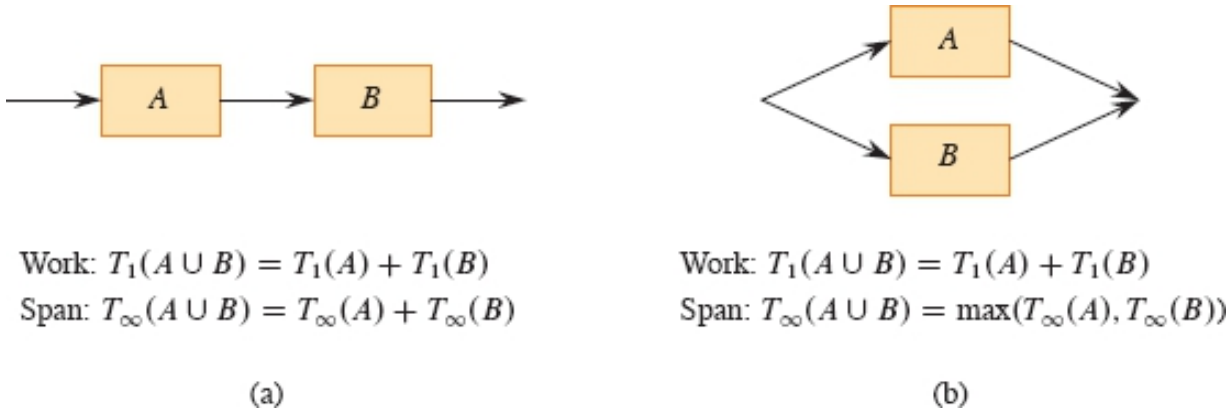


Figure 26.3 Series-parallel composition of parallel traces. **(a)** When two traces are joined in series, the work of the composition is the sum of their work, and the span of the composition is the sum of their spans. **(b)** When two traces are joined in parallel, the work of the composition remains the sum of their work, but the span of the composition is only the maximum of their spans.

Armed with an understanding of series-parallel composition, we can analyze the span of P-FIB (n). The spawned call to P-FIB ($n - 1$) in line 3 runs in parallel with the call to P-FIB ($n - 2$) in line 4. Hence, we can express the span of P-FIB (n) as the recurrence

$$\begin{aligned} T_\infty(n) &= \max\{T_\infty(n-1), T_\infty(n-2)\} + \Theta(1) \\ &= T_\infty(n-1) + \Theta(1), \end{aligned}$$

which has solution $T_\infty(n) = \Theta(n)$. (The second equality above follows from the first because P-FIB ($n - 1$) uses P-FIB ($n - 2$) in its computation, so that the span of P-FIB ($n - 1$) must be at least as large as the span of P-FIB ($n - 2$).)

The parallelism of P-FIB (n) is $T_1(n)/T_\infty(n) = \Theta(\phi^n/n)$, which grows dramatically as n gets large. Thus, Corollary 26.3 tells us that on even the largest parallel computers, a modest value for n suffices to achieve near perfect linear speedup for P-FIB (n), because this procedure exhibits considerable parallel slackness.

Parallel loops

Many algorithms contain loops for which all the iterations can operate in parallel. Although the **spawn** and **sync** keywords can be used to parallelize such loops, it is more convenient to specify directly that the iterations of such loops can run in parallel. Our pseudocode provides this functionality via the **parallel** keyword, which precedes the **for** keyword in a **for** loop statement.

As an example, consider the problem of multiplying a square $n \times n$ matrix $A = (a_{ij})$ by an n-vector $x = (x_j)$. The resulting n-vector $y = (y_i)$ is given by the equation

$$y_i = \sum_{j=1}^{n} a_{ij} x_j \ ,$$

for $i = 1, 2, \ldots, n$. The P-MAT-VEC procedure performs matrix-vector multiplication (actually, $y = y + Ax$) by computing all the entries of y in parallel. The **parallel for** keywords in line 1 of P-MAT-VEC indicate that the n iterations of the loop body, which includes a serial **for** loop, may be run in parallel. The initialization $y = 0$, if desired, should be performed before calling the procedure (and can be done with a **parallel for** loop).

```
P-MAT-VEC (A, x, y, n)

1 parallel for i = 1 to n          // parallel loop
2     for j = 1 to n               // serial loop
3         y_i = y_i + a_ij x_j
```

Compilers for fork-join parallel programs can implement **parallel for** loops in terms of **spawn** and **sync** by using recursive spawning. For example, for the **parallel for** loop in lines 1–3, a compiler can generate the auxiliary subroutine P-MAT-VEC-RECURSIVE and call P-MAT-VEC-RECURSIVE (A, x, y, n, 1, n) in the place where the loop would be in the compiled code. As Figure 26.4 illustrates, this procedure recursively spawns the first half of the iterations of the loop to execute in parallel (line 5) with the second half of the iterations (line 6) and then

executes a **sync** (line 7), thereby creating a binary tree of parallel execution. Each leaf represents a base case, which is the serial **for** loop of lines 2–3.

```
P-MAT-VEC-RECURSIVE (A, x, y, n, i, i')
1 if i == i'                    // just one iteration to do?
2     for j = 1 to n            // mimic P-MAT-VEC serial loop
3         y_i = y_i + a_ij x_j
4 else mid = ⌊(i + i')/2⌋       // parallel divide-and-conquer
5     spawn P-MAT-VEC-RECURSIVE (A, x, y, n, i, mid)
6     P-MAT-VEC-RECURSIVE (A, x, y, n, mid + 1, i')
7     sync
```

To calculate the work $T_1(n)$ of P-MAT-VEC on an $n \times n$ matrix, simply compute the running time of its serial projection, which comes from replacing the **parallel for** loop in line 1 with an ordinary **for** loop. The running time of the resulting serial pseudocode is $\Theta(n^2)$, which means that $T_1(n) = \Theta(n^2)$. This analysis seems to ignore the overhead for recursive spawning in implementing the parallel loops, however. Indeed, the overhead of recursive spawning does increase the work of a parallel loop compared with that of its serial projection, but not asymptotically. To see why, observe that since the tree of recursive procedure instances is a full binary tree, the number of internal nodes is one less than the number of leaves (see Exercise B.5-3 on page 1175). Each internal node performs constant work to divide the iteration range, and each leaf corresponds to a base case, which takes at least constant time ($\Theta(n)$ time in this case). Thus, by amortizing the overhead of recursive spawning over the work of the iterations in the leaves, we see that the overall work increases by at most a constant factor.

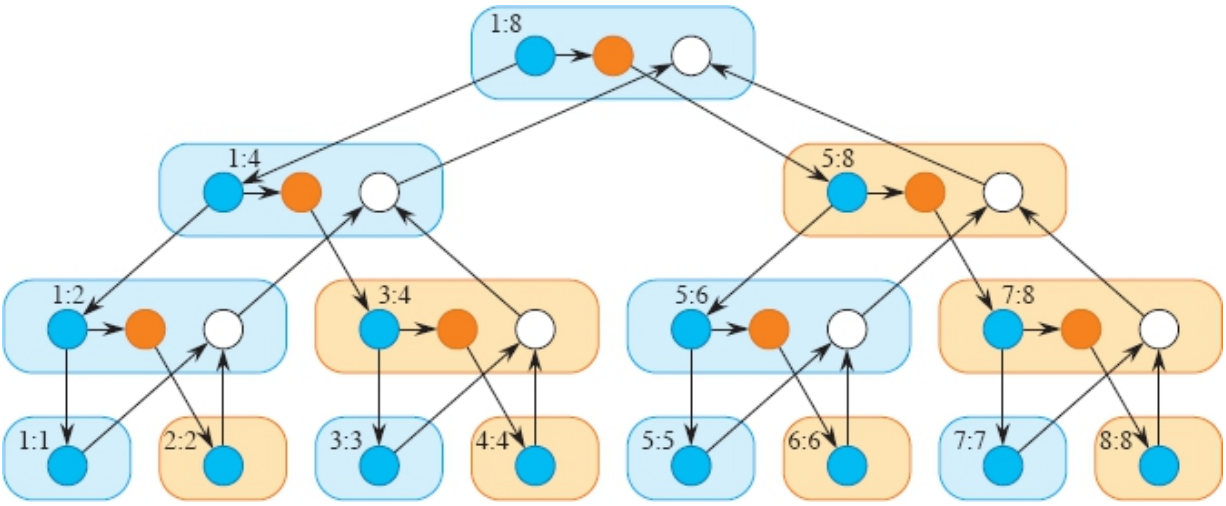


Figure 26.4 A trace for the computation of P-MAT-VEC-RECURSIVE (A, x, y, 8, 1, 8). The two numbers within each rounded rectangle give the values of the last two parameters (i and i' in the procedure header) in the invocation (spawn, in blue, or call, in tan) of the procedure. The blue circles represent strands corresponding to the part of the procedure up to the spawn of P-MAT-VEC-RECURSIVE in line 5. The orange circles represent strands corresponding to the part of the procedure that calls P-MAT-VEC-RECURSIVE in line 6 up to the **sync** in line 7, where it suspends until the spawned subroutine in line 5 returns. The white circles represent strands corresponding to the (negligible) part of the procedure after the **sync** up to the point where it returns.

To reduce the overhead of recursive spawning, task-parallel platforms sometimes ***coarsen*** the leaves of the recursion by executing several iterations in a single leaf, either automatically or under programmer control. This optimization comes at the expense of reducing the parallelism. If the computation has sufficient parallel slackness, however, near-perfect linear speedup won't be sacrificed.

Although recursive spawning doesn't affect the work of a parallel loop asymptotically, we must take it into account when analyzing the span. Consider a parallel loop with n iterations in which the ith iteration has span $iter_\infty(i)$. Since the depth of recursion is logarithmic in the number of iterations, the parallel loop's span is

$$T_\infty(n) = \Theta(\lg n) + \max\{iter_\infty(i) : 1 \le i \le n\}.$$

For example, let's compute the span of the doubly nested loops in lines 1–3 of P-MAT-VEC. The span for the **parallel for** loop control is

$\Theta(\lg n)$. For each iteration of the outer parallel loop, the inner serial **for** loop contains n iterations of line 3. Since each iteration takes constant time, the total span for the inner serial **for** loop is $\Theta(n)$, no matter which iteration of the outer **parallel for** loop it's in. Thus, taking the maximum over all iterations of the outer loop and adding in the $\Theta(\lg n)$ for loop control yields an overall span of $T_\infty n = \Theta(n) + \Theta(\lg n) = \Theta(n)$ for the procedure. Since the work is $\Theta(n^2)$, the parallelism is $\Theta(n^2)/\Theta(n) = \Theta(n)$. (Exercise 26.1-7 asks you to provide an implementation with even more parallelism.)

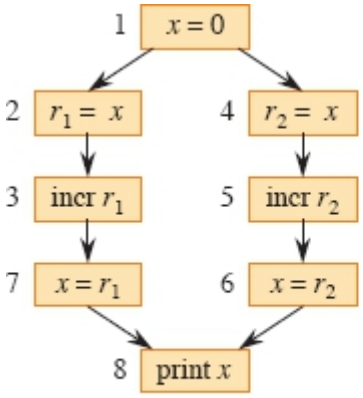
Race conditions

A parallel algorithm is ***deterministic*** if it always does the same thing on the same input, no matter how the instructions are scheduled on the multicore computer. It is ***nondeterministic*** if its behavior might vary from run to run when the input is the same. A parallel algorithm that is intended to be deterministic may nevertheless act nondeterministically, however, if it contains a difficult-to-diagnose bug called a "determinacy race."

Famous race bugs include the Therac-25 radiation therapy machine, which killed three people and injured several others, and the Northeast Blackout of 2003, which left over 50 million people in the United States without power. These pernicious bugs are notoriously hard to find. You can run tests in the lab for days without a failure, only to discover that your software sporadically crashes in the field, sometimes with dire consequences.

A ***determinacy race*** occurs when two logically parallel instructions access the same memory location and at least one of the instructions modifies the value stored in the location. The toy procedure RACE-EXAMPLE on the following page illustrates a determinacy race. After initializing x to 0 in line 1, RACE-EXAMPLE creates two parallel strands, each of which increments x in line 3. Although it might seem that a call of RACE-EXAMPLE should always print the value 2 (its serial projection certainly does), it could instead print the value 1. Let's see how this anomaly might occur.

When a processor increments x, the operation is not indivisible, but is composed of a sequence of instructions:



(a)

step	x	r_1	r_2
1	0	–	–
2	0	0	–
3	0	1	–
4	0	1	0
5	0	1	1
6	1	1	1
7	1	1	1

(b)

Figure 26.5 Illustration of the determinacy race in RACE-EXAMPLE. **(a)** A trace showing the dependencies among individual instructions. The processor registers are r_1 and r_2. Instructions unrelated to the race, such as the implementation of loop control, are omitted. **(b)** An execution sequence that elicits the bug, showing the values of x in memory and registers r_1 and r_2 for each step in the execution sequence.

```
RACE-EXAMPLE ( )
1  x = 0
2  parallel for i = 1 to 2
3      x = x + 1                    // determinacy race
4  print x
```

- Load x from memory into one of the processor's registers.
- Increment the value in the register.
- Store the value in the register back into x in memory.

Figure 26.5(a) illustrates a trace representing the execution of RACE-EXAMPLE, with the strands broken down to individual instructions. Recall that since an ideal parallel computer supports sequential consistency, you can view the parallel execution of a parallel algorithm as an interleaving of instructions that respects the dependencies in the

trace. Part (b) of the figure shows the values in an execution of the computation that elicits the anomaly. The value x is kept in memory, and r_1 and r_2 are processor registers. In step 1, one of the processors sets x to 0. In steps 2 and 3, processor 1 loads x from memory into its register r_1 and increments it, producing the value 1 in r_1. At that point, processor 2 comes into the picture, executing instructions 4–6. Processor 2 loads x from memory into register r_2; increments it, producing the value 1 in r_2; and then stores this value into x, setting x to 1. Now, processor 1 resumes with step 7, storing the value 1 in r_1 into x, which leaves the value of x unchanged. Therefore, step 8 prints the value 1, rather than the value 2 that the serial projection would print.

Let's recap what happened. By sequential consistency, the effect of the parallel execution is as if the executed instructions of the two processors are interleaved. If processor 1 executes all its instructions before processor 2, a trivial interleaving, the value 2 is printed. Conversely, if processor 2 executes all its instructions before processor 1, the value 2 is still printed. When the instructions of the two processors interleave nontrivially, however, it is possible, as in this example execution, that one of the updates to x is lost, resulting in the value 1 being printed.

Of course, many executions do not elicit the bug. That's the problem with determinacy races. Generally, most instruction orderings produce correct results, such as any where the instructions on the left branch execute before the instructions on the right branch, or vice versa. But some orderings generate improper results when the instructions interleave. Consequently, races can be extremely hard to test for. Your program may fail, but you may be unable to reliably reproduce the failure in subsequent tests, confounding your attempts to locate the bug in your code and fix it. Task-parallel programming environments often provide race-detection productivity tools to help you isolate race bugs.

Many parallel programs in the real world are intentionally nondeterministic. They contain determinacy races, but they mitigate the dangers of nondeterminism through the use of mutual-exclusion locks and other methods of synchronization. For our purposes, however, we'll

insist on an absence of determinacy races in the algorithms we develop. Nondeterministic programs are indeed interesting, but nondeterministic programming is a more advanced topic and unnecessary for a wide swath of interesting parallel algorithms.

To ensure that algorithms are deterministic, any two strands that operate in parallel should be ***mutually noninterfering***: they only read, and do not modify, any memory locations accessed by both of them. Consequently, in a **parallel for** construct, such as the outer loop of P-MAT-VEC, we want all the iterations of the body, including any code an iteration executes in subroutines, to be mutually noninterfering. And between a **spawn** and its corresponding **sync**, we want the code executed by the spawned child and the code executed by the parent to be mutually noninterfering, once again including invoked subroutines.

As an example of how easy it is to write code with unintentional races, the P-MAT-VEC-WRONG procedure on the next page is a faulty parallel implementation of matrix-vector multiplication that achieves a span of $\Theta(\lg n)$ by parallelizing the inner **for** loop. This procedure is incorrect, unfortunately, due to determinacy races when updating y_i in line 3, which executes in parallel for all n values of j.

Index variables of **parallel for** loops, such as i in line 1 and j in line 2, do not cause races between iterations. Conceptually, each iteration of the loop creates an independent variable to hold the index of that iteration during that iteration's execution of the loop body. Even if two parallel iterations both access the same index variable, they really are accessing different variable instances—hence different memory locations—and no race occurs.

```
P-MAT-VEC-WRONG (A, x, y, n)
1 parallel for i = 1 to n
2     parallel for j = 1 to n
3         y_i = y_i + a_ij x_j          // determinacy race
```

A parallel algorithm with races can sometimes be deterministic. As an example, two parallel threads might store the same value into a

shared variable, and it wouldn't matter which stored the value first. For simplicity, however, we generally prefer code without determinacy races, even if the races are benign. And good parallel programmers frown on code with determinacy races that cause nondeterministic behavior, if deterministic code that performs comparably is an option.

But nondeterministic code does have its place. For example, you can't implement a parallel hash table, a highly practical data structure, without writing code containing determinacy races. Much research has centered around how to extend the fork-join model to incorporate limited "structured" nondeterminism while avoiding the full measure of complications that arise when nondeterminism is completely unrestricted.

A chess lesson

To illustrate the power of work/span analysis, this section closes with a true story that occurred during the development of one of the first world-class parallel chess-playing programs [106] many years ago. The timings below have been simplified for exposition.

The chess program was developed and tested on a 32-processor computer, but it was designed to run on a supercomputer with 512 processors. Since the supercomputer availability was limited and expensive, the developers ran benchmarks on the small computer and extrapolated performance to the large computer.

At one point, the developers incorporated an optimization into the program that reduced its running time on an important benchmark on the small machine from $T_{32} = 65$ seconds to $T'_{32} = 40$ seconds. Yet, the developers used the work and span performance measures to conclude that the optimized version, which was faster on 32 processors, would actually be slower than the original version on the 512 processors of the large machine. As a result, they abandoned the "optimization."

Here is their work/span analysis. The original version of the program had work $T_1 = 2048$ seconds and span $T_\infty = 1$ second. Let's treat inequality (26.4) on page 760 as the equation $T_P = T_1/P + T_\infty$, which we can use as an approximation to the running time on P processors.

Then indeed we have $T_{32} = 2048/32 + 1 = 65$. With the optimization, the work becomes $T'_1 = 1024$ seconds, and the span becomes $T'_\infty = 8$ seconds. Our approximation gives $T'_{32} = 1024/32 + 8 = 40$.

The relative speeds of the two versions switch when we estimate their running times on 512 processors, however. The first version has a running time of $T_{512} = 2048/512+1 = 5$ seconds, and the second version runs in $T'_{512} = 1024/512+8 = 10$ seconds. The optimization that speeds up the program on 32 processors makes the program run for twice as long on 512 processors! The optimized version's span of 8, which is not the dominant term in the running time on 32 processors, becomes the dominant term on 512 processors, nullifying the advantage from using more processors. The optimization does not scale up.

The moral of the story is that work/span analysis, and measurements of work and span, can be superior to measured running times alone in extrapolating an algorithm's scalability.

Exercises

26.1-1

What does a trace for the execution of a serial algorithm look like?

26.1-2

Suppose that line 4 of P-FIB spawns P-FIB ($n - 2$), rather than calling it as is done in the pseudocode. How would the trace of P-FIB(4) in Figure 26.2 change? What is the impact on the asymptotic work, span, and parallelism?

26.1-3

Draw the trace that results from executing P-FIB(5). Assuming that each strand in the computation takes unit time, what are the work, span, and parallelism of the computation? Show how to schedule the trace on 3 processors using greedy scheduling by labeling each strand with the time step in which it is executed.

26.1-4

Prove that a greedy scheduler achieves the following time bound, which is slightly stronger than the bound proved in Theorem 26.1:

$$T_P \le \frac{T_1 - T_\infty}{P} + T_\infty \,. \tag{26.5}$$

26.1-5

Construct a trace for which one execution by a greedy scheduler can take nearly twice the time of another execution by a greedy scheduler on the same number of processors. Describe how the two executions would proceed.

26.1-6

Professor Karan measures her deterministic task-parallel algorithm on 4, 10, and 64 processors of an ideal parallel computer using a greedy scheduler. She claims that the three runs yielded $T_4 = 80$ seconds, $T_{10} = 42$ seconds, and $T_{64} = 10$ seconds. Argue that the professor is either lying or incompetent. (*Hint:* Use the work law (26.2), the span law (26.3), and inequality (26.5) from Exercise 26.1-4.)

26.1-7

Give a parallel algorithm to multiply an $n \times n$ matrix by an n-vector that achieves $\Theta(n^2/\lg n)$ parallelism while maintaining $\Theta(n^2)$ work.

26.1-8

Analyze the work, span, and parallelism of the procedure P-TRANSPOSE, which transposes an $n \times n$ matrix A in place.

```
P-TRANSPOSE (A, n)
1  parallel for j = 2 to n
2      parallel for i = 1 to j − 1
3          exchange a_ij with a_ji
```

26.1-9

Suppose that instead of a **parallel for** loop in line 2, the P-TRANSPOSE procedure in Exercise 26.1-8 had an ordinary **for** loop. Analyze the

work, span, and parallelism of the resulting algorithm.

26.1-10
For what number of processors do the two versions of the chess program run equally fast, assuming that $T_P = T_1/P + T_\infty$?

26.2 Parallel matrix multiplication

In this section, we'll explore how to parallelize the three matrix-multiplication algorithms from Sections 4.1 and 4.2. We'll see that each algorithm can be parallelized in a straightforward fashion using either parallel loops or recursive spawning. We'll analyze them using work/span analysis, and we'll see that each parallel algorithm attains the same performance on one processor as its corresponding serial algorithm, while scaling up to large numbers of processors.

A parallel algorithm for matrix multiplication using parallel loops

The first algorithm we'll study is P-MATRIX-MULTIPLY, which simply parallelizes the two outer loops in the procedure MATRIX-MULTIPLY on page 81.

```
P-MATRIX-MULTIPLY (A, B, C, n)
1  parallel for i = 1 to n          // compute entries in each of n rows
2      parallel for j = 1 to n      // compute n entries in row i
3          for k = 1 to n
4              c_ij = c_ij + a_ik · b_kj   // add in another term of equation (4.1)
```

Let's analyze P-MATRIX-MULTIPLY. Since the serial projection of the algorithm is just MATRIX-MULTIPLY, the work is the same as the running time of MATRIX-MULTIPLY: $T_1(n) = \Theta(n^3)$. The span is $T_\infty(n) = \Theta(n)$, because it follows a path down the tree of recursion for the **parallel for** loop starting in line 1, then down the tree of recursion